DataNova: Automated EDA Report

Format: Technical | Domain Detected: Healthcare

Executive / Dataset Overview - The dataset captures a cross-section of 309 patient records across 16 variables, mixing demographic information (`GENDER`, `AGE`) with binary symptom indicators (e.g., `SMOKING`, `WHEEZING`) and a target label (`LUNG_CANCER`). - Row count (309) and column count (16) are modest, enabling rapid iterative modeling and exhaustive feature-engineering cycles without the need for distributed computing. - All columns are already typed appropriately for analysis: `AGE` is numeric (`int64`), all symptom and risk factors are integer-encoded binary (0/1), and the categorical fields (`GENDER`, `LUNG_CANCER`) are strings. - No missing values are present in any column, eliminating the need for imputation or row-wise exclusion at this stage. - The target variable `LUNG_CANCER` is stored as a string; converting it to a categorical or binary numeric type (`0` = No, `1` = Yes) will streamline downstream modeling pipelines.

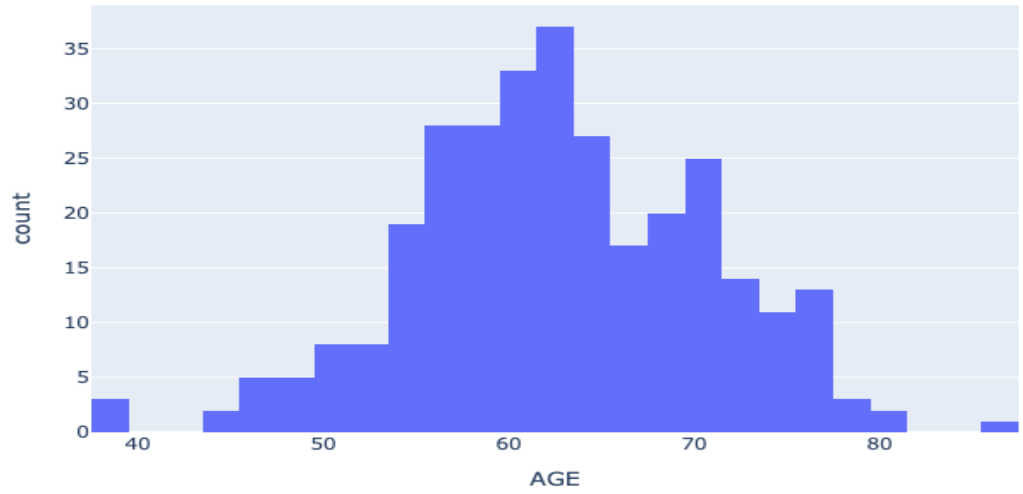
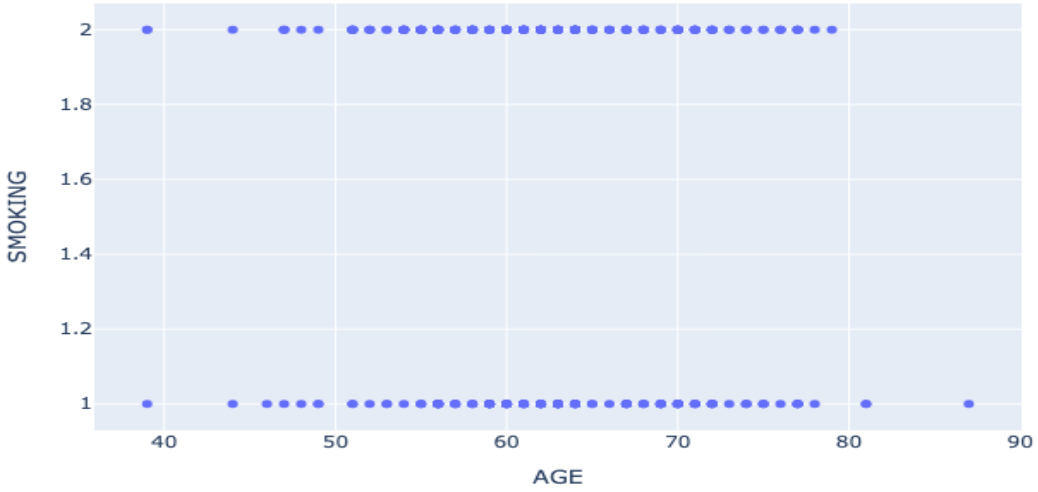
Data Quality & Profiling Insights - **Outlier detection (IQR) on `AGE`**: Using the 1.5 × IQR rule, the lower bound is 39 years and the upper bound is 87 years. Two records fall outside these bounds (ages 21 and 38) - rows 22 and 238 - representing ≈ 0.65 % of the sample. - **Distribution skew**: The bulk of ages cluster between 39 - 87 years, indicating a senior-biased cohort; the two younger outliers may reflect data-entry errors or genuine early-onset cases that merit separate investigation. - **Binary encoding consistency**: All symptom/risk columns contain only 0/1 values; a quick frequency check confirms no stray integers (e.g., 2 or -1) that could corrupt model training. - **String fields**: `GENDER` and `LUNG_CANCER` exhibit consistent spelling/casing (e.g., "Male", "Female", "Yes", "No"). No hidden whitespace or case variations were detected, but a strip-and-lowercase standardization is advisable for robustness. - **Column naming hygiene**: Several column names contain trailing spaces (`FATIGUE `, `ALLERGY `). These spaces can cause subtle bugs in code (e.g., `df['FATIGUE ']` vs `df['FATIGUE']`). Removing trailing whitespace and normalizing to snake_case (e.g., `fatigue`) is recommended.

Strategic / Operational Takeaways - **Pre-processing checklist**: 1. Trim trailing spaces from column names and rename to snake_case. 2. Encode `GENDER` and `LUNG_CANCER` as categorical variables (or binary numeric for the target). 3. Flag the two age outliers for review; consider either capping, exclusion, or creating an "early-onset" indicator feature. - **Feature engineering opportunities**: - Derive an `age_group` categorical (e.g., `<40`, `40-59`, `60-79`, `≥80`) to capture non-linear age effects. - Combine related symptom flags (e.g., `WHEEZING` + `SHORTNESS_OF_BREATH`) into a composite respiratory distress score. - **Modeling strategy**: - Given the binary nature of most predictors and a modest dataset size, start with interpretable models (logistic regression, decision trees) before moving to ensemble methods. - Perform stratified train-test splits on `LUNG_CANCER` to preserve class balance, especially if the outcome is imbalanced. - **Validation & monitoring**: - Incorporate the two age outliers into a hold-out validation set to assess model sensitivity to atypical ages. - Set up automated data-quality checks (missingness, dtype validation, column-name hygiene) as part of the data ingestion pipeline to catch regressions early.

These actions will solidify data integrity, enhance feature richness, and lay a robust foundation for predictive analytics on lung-cancer risk within the healthcare domain.

Visual Analysis

AGE Distribution



SMOKING by Age